

1. NAME OF THE MEDICINAL PRODUCT

PROSTIN E2® 3 mg Vaginal Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Active ingredient: dinoprostone as the naturally occurring prostaglandin E₂ (PGE₂). Vaginal tablets containing 3 mg dinoprostone.

3. PHARMACEUTICAL FORM

Tablet for vaginal administration.

A white tablet marked "UPJOHN" and "715" on one side, plain on the other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Oxytocic agent. Dinoprostone are indicated for the induction of labour, especially in patients with favourable induction features, when there are no foetal or maternal contraindications.

4.2 Posology and method of administration

The initial dose is 1 tablet (3 mg) of dinoprostone inserted high into the posterior fornix. A second tablet may be inserted after 6 – 8 hours if labor has not been established. The maximum or total dose in 24 hours is 6 mg.

Usage is restricted to qualified health care professionals and to hospitals and clinics with specialised obstetric units with facilities for continuous monitoring.

The recommended dose should not be exceeded, and the dosing interval should not be shortened as this increases the risk of uterine hyperstimulation, uterine rupture, uterine haemorrhage, foetal and neonatal death.

4.3 Contraindications

Hypersensitivity to the active substance(s) or to any of the excipients listed in section 6.1.

Dinoprostone should not be used where the patient is sensitive to prostaglandins or other constituents of the tablet.

Dinoprostone are not recommended in the following circumstances:

- For patients in whom oxytocic drugs are generally contraindicated or where prolonged contractions of the uterus are considered inappropriate, such as:
 - Cases with a history of Caesarean section or major uterine surgery
 - Cases where there is cephalopelvic disproportion
 - Cases in which foetal malpresentation is present
 - Cases where there is clinical suspicion or definite evidence of pre-existing foetal distress
 - Cases in which there is a history of difficult labour and/or traumatic delivery
- Unexplained vaginal discharge and/or abnormal uterine bleeding during current pregnancy.
- In patients with a past history of, or existing, pelvic inflammatory disease, unless adequate

prior treatment has been instituted.

- In patients where there is clinical suspicion or definite evidence of placenta praevia or unexplained vaginal bleeding during this pregnancy.
- Patients with active cardiac, pulmonary, renal or hepatic disease.

4.4 Special warnings and precautions for use

This product is only available to hospitals and clinics with specialised obstetric units and should only be used where 24-hour resident medical cover is provided.

Use caution in handling this product to prevent contact with skin. Wash hands thoroughly with soap and water after administration.

As with any oxytocic agent, the risk of uterine rupture should be considered. Concomitant medication, maternal and foetal status should be taken into consideration in order to minimise the risk of uterine hyperstimulation, uterine rupture, uterine haemorrhage, foetal and neonatal death. Careful and regular monitoring of uterine activity and foetal heart rate should be conducted during use of dinoprostone. Patients who develop uterine hypertonus or hypercontractility, or in whom unusual foetal heart rate patterns develop, should be managed in a manner that addresses the welfare of the foetus and mother.

Caution should be exercised in the administration of dinoprostone for the induction of labour in patients with:

- asthma or a history of asthma
- epilepsy or a history of epilepsy
- glaucoma or raised intra-ocular pressure
- compromised cardiovascular, hepatic, or renal function
- hypertension
- ruptured chorioamniotic membranes

Dinoprostone should be used with caution in patients with multiple pregnancy.

In labour induction, cephalopelvic relationships should be carefully evaluated before use of dinoprostone. During use, uterine activity, foetal status and the progression of cervical dilation should be carefully monitored to detect possible evidence of undesired responses, e.g. hypertonus, sustained uterine contractions, or foetal distress.

In cases where there is a known history of hypertonic uterine contractility or tetanic uterine contractions, it is recommended that uterine activity and the state of the foetus (where applicable) should be continuously monitored throughout labour. The possibility of uterine rupture should be borne in mind where high-tone uterine contractions are sustained.

Women aged 35 years or older, those with complications during pregnancy and those with a gestational age over 40 weeks have been shown to have an increased risk of post-partum disseminated intravascular coagulation. In addition, these factors may further increase the risk associated with labor induction (see section **4.8 Undesirable effects**). Therefore, in these women, use of dinoprostone should be undertaken with caution. Measures should be applied to detect as soon as possible an evolving fibrinolysis in the immediate post-partum phase. The clinician should be alert that the intracervical placement of dinoprostone may result in inadvertent disruption and subsequent embolization of antigenic tissue, causing in rare circumstances the development of Anaphylactoid Syndrome of Pregnancy (Amniotic Fluid Embolism).

4.5 Interaction with other medicinal products and other forms of interaction

The response to oxytocin may be accentuated in the presence of exogenous prostaglandin therapy. Concurrent use with other oxytocic agents is not recommended. The sequential use of oxytocin following administration of vaginal tablets is recommended, with a dosing interval of at least 6 hours.

4.6 Pregnancy and lactation

Pregnancy

Dinoprostone is for use in pregnant women at or near term.

Prostaglandin E2 produced an increase in skeletal anomalies in rats and rabbits. Dinoprostone has been shown to be embryotoxic in rats and rabbits, and any dose that produces sustained increased uterine tone could put the embryo or fetus at risk (see section **4.4 Special warnings and precautions for use**).

Lactation

Prostaglandins are excreted in breast milk at very low concentrations. No measurable differences were observed in the milk of mothers delivering prematurely and at term.

4.7 Effects on ability to drive and use machines

Not applicable

4.8 Undesirable effects

Maternal Adverse Events

The following maternal adverse events have been reported with use of vaginal tablets:

Immune system disorders: Hypersensitivity reactions (e.g., Anaphylactic reaction, Anaphylactic shock, Anaphylactoid reaction)

Gastrointestinal disorders: Diarrhea, nausea, vomiting

Musculoskeletal and connective tissue disorders: Back pain

Pregnancy, puerperium and perinatal conditions: Uterine contractile abnormalities (increase frequency, tone, or duration), uterine rupture, abruptio placenta, pulmonary amniotic fluid embolism, rapid cervical dilatation

Reproductive system and breast disorders: Warm feeling in vagina

General disorders and administration site conditions: Fever

Vascular disorders: Hypertension

Respiratory, thoracic and mediastinal disorders: Asthma, bronchospasm

Fetal Adverse Events

The following fetal adverse events have been reported with use of vaginal tablets:

Pregnancy, puerperium and perinatal conditions: Foetal death, Still births, Neonatal death

Investigations: Fetal distress/altered fetal heart rate (FHR), neonatal distress/low Apgar score

Foetal death, stillbirth, and neonatal death have been reported after application of dinoprostone, especially following the occurrence of serious events such as uterine rupture (**see sections 4.2, 4.3 and 4.4**).

Post-marketing surveillance

Blood and lymphatic system disorders: An increased risk of post-partum disseminated intravascular coagulation has been described in patients whose labor was induced by pharmacological means, either with dinoprostone or oxytocin (see section **4.4 Special warnings and precautions for use**). The frequency of this adverse event, however, appears to be rare (<1 per 1,000 labors).

4.9 Overdose

Overdosage may be expressed by uterine hypercontractility and uterine hypertonus. Because of the transient nature of PGE₂-induced myometrial hyperstimulation, nonspecific, conservative management was found to be effective in the vast majority of the cases; i.e., maternal position change and administration of oxygen to the mother. B-adrenergic drugs may be used as a treatment of hyperstimulation following administration of PGE₂ for cervical ripening.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Mechanism of Action/Effect

For uterine stimulation

Dinoprostone stimulates the myometrium of the gravid uterus to contract in a manner that is similar to the contractions seen in the term uterus during labor. Whether or not this action results from a direct effect of dinoprostone on the myometrium has not been determined. Nonetheless, the myometrial contractions induced by the vaginal administration of dinoprostone are sufficient to produce evacuation of the products of conception from the uterus in the majority of cases.

For cervical ripening

Dinoprostone has a local cervical effect in initiating softening, effacement, and dilation. These changes, referred to as cervical ripening, occur spontaneously as the normal pregnancy progresses toward term and allow evacuation of uterine contents by decreasing cervical resistance at the same time that myometrial activity increases.

Other actions

Dinoprostone is also capable of stimulating smooth muscle of the gastrointestinal tract in humans. This activity may be responsible for the vomiting and/or diarrhea that is occasionally seen when dinoprostone is used for preinduction cervical ripening.

In laboratory animals, and also in humans, large doses of dinoprostone can lower blood pressure, probably as a result of its effect on smooth muscle of the vascular system. Dinoprostone can also elevate body temperature; however, with the dose of dinoprostone used for cervical ripening, these effects have not been seen.

5.2 Pharmacokinetic properties

General characteristics of active substances

Absorption

When administered vaginally, dinoprostone is rapidly absorbed. Dinoprostone is 73% bound to human plasma albumin.

Following insertion of the vaginal tablet, PGE₂ absorption (as measured by the presence of PGE₂ metabolites) increases to reach a peak at about 40 minutes.

Distribution and Metabolism

Dinoprostone is widely distributed in the mother.

PGE₂ is rapidly metabolized to 13, 14-dihydro-15-keto PGE₂, which is converted to 13, 14-dihydro, 15-keto PGA₂. Dinoprostone is completely metabolized in humans. It is extensively metabolized in the lungs, and the resulting metabolites are further metabolized in the liver and kidney.

Elimination

The drug and its metabolites are excreted primarily by the kidneys, with a small amount excreted in the feces.

5.3 Preclinical safety data

Carcinogenesis, mutagenesis, impairment of fertility

Carcinogenic bioassay studies have not been conducted in animals with dinoprostone due to the limited indications for use and short duration of administration. No evidence of mutagenicity was observed in the Micronucleus Test or Ames Assay.

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Lactose

Microcrystalline Cellulose Colloidal Silicon Dioxide Maize Starch

Magnesium Stearate

6.2 Incompatibilities

Not applicable.

6.3 Shelf life

Please see expiry date on the outer carton.

6.4 Special precautions for storage

Store in the refrigerator (2°C – 8°C).

The tablets should be used within 1 month of opening the foil.

6.5 Nature and contents of container

Aluminium foil strip of 4 tablets, each box containing 4 or 8 tablets. Not all pack sizes may be marketed.

6.6 Special precautions for disposal and other handling

Wash hands thoroughly with soap and water after administration.

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. MANUFACTURER

Sanico N.V.
Veedijk 59,
Turnhout, 2300,
Belgium.

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